

Cover letter — DRAFT, NOT SENT

[AUTHOR LIST AND SIGNATURE — HUMAN ACTION REQUIRED]

To the Editors,

We submit *An exact degree-ten classification of local invariants of the ten-dimensional self-dual five-form* for consideration.

A self-dual five-form in ten dimensions admits 81 functionally independent Lorentz invariants. That count is analytic and known. What has not been available is the structure of the invariant ring at fixed degree, and consequently there has been no way to decide whether a given local term is new, whether it is reachable by the stress-tensor flow constructions that motivate these theories, or how a proposed deformation relates to the explicit structures already in the literature.

We determine the degree-ten graded piece exactly. It is fourteen-dimensional. We resolve its internal structure completely — products, published span, graph generators, and the subspace reachable by the stress flow — with all pairwise intersections certified over finite fields at two primes. The quotient by the reachable subspace is three-dimensional and we give explicit representatives, so the deficit of the flow construction at degree ten is now a concrete, presentable object rather than an unknown.

Both dimensions that a finite-field computation could not settle are exact over the rationals. The first is the one just quoted. The reachable subspace is the one object here not given by a spanning set — it is assembled by a closure that admits a direction when it raises the rank modulo p — so its dimension was initially only a lower bound, leaving the quotient bounded above by three. Re-running the closure in exact rational arithmetic, after lifting its non-integral flow targets by Chinese remaindering and rational reconstruction validated against a held-out prime, gives eleven exactly, with an explicit non-vanishing integer minor as witness. The quotient is therefore three over $\mathbb{Q}$.

The second is the intersection of the published span with the products, and it is the one we care most about, since it carries the correction below. Lifting the published coordinates the same way gives intersection dimension exactly one, and the intersection is generated by an explicit integer identity: a specific combination of four of the twelve published structures is a multiple of the product of the degree-four and degree-six invariants. We verified that identity at a prime and on five-forms that took no part in constructing it.

One result is a correction. Two natural decompositions of the fourteen-dimensional space both read twelve plus two, and it is tempting to identify them. They are not the same: the published span contains one product direction, so its non-product content is eleven rather than twelve. We found this by computing it after nearly asserting the opposite, and we report it prominently because a reader who assumes otherwise will miscount by one.

We also connect the tensor description to a spinor-variable description through an explicitly convention-controlled bridge, verified to be an isomorphism onto the gamma-traceless symmetric chiral square with a certified left inverse, and we compare the two descriptions at a common registry of five-forms. At degree ten the spans coincide. At degree eight they do not, and we report the shortfall as a property of the candidate family rather than of the correspondence.

Finally, we replace a floating-point Jacobian rank computation, which we show to be genuinely fragile, by an exact modular one that yields a rigorous bound.

We have been careful about what is not claimed. The number 81 is analytic and is not ours. The reverse search is bounded and is not exhaustive. No generating set for the ring is presented, degree twelve is not classified, and no physical prediction is derived. These limits are collected in a dedicated section.

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